Healthcare Analysis
DIY Project 2

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Problem Statement of Healthcare Analytics

The purpose of the Healthcare Analysis project is to investigate health care visits based on demographics, financial variables and health conditions. The goal is to find patterns or trends between variable factors such as gender, age, income, illness, a reduction in activities, health condition, private insurance status, free health care support and long-term sickness.

Project Description of Healthcare Analysis

The project aims at conducting an analysis on a Healthcare Dataset so as to generate useful insights and derive meaningful inference regarding utilization of healthcare, individuals' health, illnesses and disease patterns, and influencing factors that affect number of healthcare visits. The dataset contain data of an individual falling in several criteria such as Gender, Age, Income, illness, reduced days by health condition, health, health insured and chronic health conditions.

This is an end-to-end Data Analysis project of Healthcare Data, using Python. It comprises of Data Cleaning, exploratory Data Analysis (EDA), Statistical Analysis, and Data Visualization in order to derive some interesting and useful insights from the available Data.

The analysis aims at determining relationship between factors related to demographic and healthcare factors and Healthcare Utilization. Thus helping understand how age, gender, income, illness, condition of health, conditions such as chronic disease affect the count of Healthcare visits.

Dataset Fields Given are few of the important fields of the Dataset on Healthcare that will be used in this analysis:

S. No Unique serial number of the records.

Visits Number of doctor or hospital visits by a person.

Gender of a person

Age of a person.

Income of a person.

Illness number of the conditions or level of illness of a person.

Reduced days for which a person had been subjected to reduced activity due to Illness condition.

Health of a person.

Private indicator whether a person has private health insurance or not

Free poor Indicator whether a person is qualified to have the free service for his/her to get healthcare.

Free repat Indicator whether a person is qualified to have the free service for their old age/pensioner.

Nchronic Number of chronic diseases the person have had.

Lchronic Indicator of having serious, non-transient illnesses or diseases.

Project Objectives

1) Display the First 15 Rows

2) Display complete information about the columns of the dataset such as column name,Count,Data type and overall memory usage

3) Find out the total no of people based on their count of illness

4) Visualize and analyze the maximum,mimimum and medium income

5) Find out the no of days of reduced activity of male and female separately due to illness

6) Visualize is there is any missing values in the dataset based on a heatmap

7) Find out the correlation between variables in the given dataset correlation between different variables

8) Analyze how the income of a patient affects the no of visits to the hospital

9) Count and visualize the number of males and females affected by illness

10) Visualize the percentage of people getting govt health insurance due to low income, due to old age and also the percentage of people having private health insurance

11) Plot a horizontal bar chart to analyze the reduced days of activity due to illness based on Gender

What is Data Wrangling ?

Data wrangling, a crucial phase in the data analysis process, involves preparing and cleaning raw data to make it suitable for analysis. This involves various tasks such as handling missing values, removing duplicates, and transforming data into a consistent format. Effective data wrangling ensures that the data is reliable and ready for exploration, enabling meaningful insights to be derived from it.

What is the Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)is a vital phase in data analysis involving the examination and visualization of data to uncover patterns ,trends, and insights.in the context of an Airbnb data analysis project, EDA allows researchers to gain a Comprehensive understanding of the dataset's characteristics, distribution and relationships between variables. Through techniques such as summary stastics, data visualization ,and correlation analysis, EDA sets the stage for further investigation and hypothesis testing, uiding subsequent analytical decisions.

End Users of this Project

Healthcare Organizations. Data used to study patient utilization patterns and make healthcare services better.

Hospitals and Clinics. Data to study pattern of visits, diseases and chronic diseases among patients so can forecast the demands properly.

Doctors and health Professionals. Data used to analyze trend related to health and identify factors involved in many visits to hospital/health center.

Government and Health Departments. To study patterns related to health and assist in policymaking, resource allocation, and health planning

Health insurance Companies. Health care utilization patterns are used to analyze and discover their relations to health condition.

Researchers and Data Analyst. Perform statistical analysis and study correlation factors between demography, economics and health related issues.

Health care manager. Supports decision making by evidence obtained from data and to improve administrative work.

Patients and Publics. The information could be used as source of understanding people's access to healthcare and the affecting variables which will influences health related problems.

Public health organizations. Data are useful to detect different health problems in society so to plan the related projects.

Students and Public organizations. Dataset is taken to illustrate and take an action with real condition, such as implementing python programming skills, explore data, statistical analysis and data visualization

Technologies Used In this Project

Hardware Requirements

The hardware requirements for performing the Healthcare Analysis are:

RAM – 8.00 GB

Operating System – Windows 11

Processor – Intel(R) Core(TM) i3-1115g4

Processor speed – 3.00 GHz

Software Requirements

The software requirements for performing the Healthcare Analysis are:

Programming Language – Python

Development Environment / IDE – Jupyter Notebook

Data Analysis Libraries – Pandas, NumPy

Data Visualization Libraries – Matplotlib, Seaborn

Dataset Format – CSV

Results of the Project

1) Display the First 15 Rows

```
[5]: df.head(15)
```

[5]:

	S.no	visits	gender	age	income	illness	reduced	health	private	freepoor	freerepat	nchronic	lchronic
0	1	1	female	0.19	0.55	1	4	1	yes	no	no	no	no
1	2	1	female	0.19	0.45	1	2	1	yes	no	no	no	no
2	3	1	male	0.19	0.90	3	0	0	no	no	no	no	no
3	4	1	male	0.19	0.15	1	0	0	no	no	no	no	no
4	5	1	male	0.19	0.45	2	5	1	no	no	no	yes	no
5	6	1	female	0.19	0.35	5	1	9	no	no	no	yes	no
6	7	1	female	0.19	0.55	4	0	2	no	no	no	no	no
7	8	1	female	0.19	0.15	3	0	6	no	no	no	no	no
8	9	1	female	0.19	0.65	2	0	5	yes	no	no	no	no
9	10	1	male	0.19	0.15	1	0	0	yes	no	no	no	no
10	11	1	male	0.19	0.45	1	0	0	no	no	no	no	no
11	12	1	male	0.19	0.25	2	0	2	no	no	yes	no	no
12	13	2	male	0.19	0.55	3	13	1	no	no	no	yes	no
13	14	1	male	0.19	0.45	4	7	6	no	no	no	yes	no
14	15	1	male	0.19	0.25	3	1	0	yes	no	no	yes	no

2) Display complete information about the columns of the dataset such as column name,Count,Data type and overall memory usage

```
[6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5190 entries, 0 to 5189
Data columns (total 13 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   S.no       5190 non-null   int64
 1   visits     5190 non-null   int64
 2   gender     5190 non-null   object
 3   age        5190 non-null   float64
 4   income     5190 non-null   float64
 5   illness    5190 non-null   int64
 6   reduced    5190 non-null   int64
 7   health     5190 non-null   int64
 8   private    5190 non-null   object
 9   freepoor   5190 non-null   object
 10  freerepat  5190 non-null   object
 11  nchronic   5190 non-null   object
 12  lchronic   5190 non-null   object
dtypes: float64(2), int64(5), object(6)
memory usage: 527.2+ KB
```

3) Find out the total no of people based on their count of illness

```
[7]: df["illness"].value_counts()
```

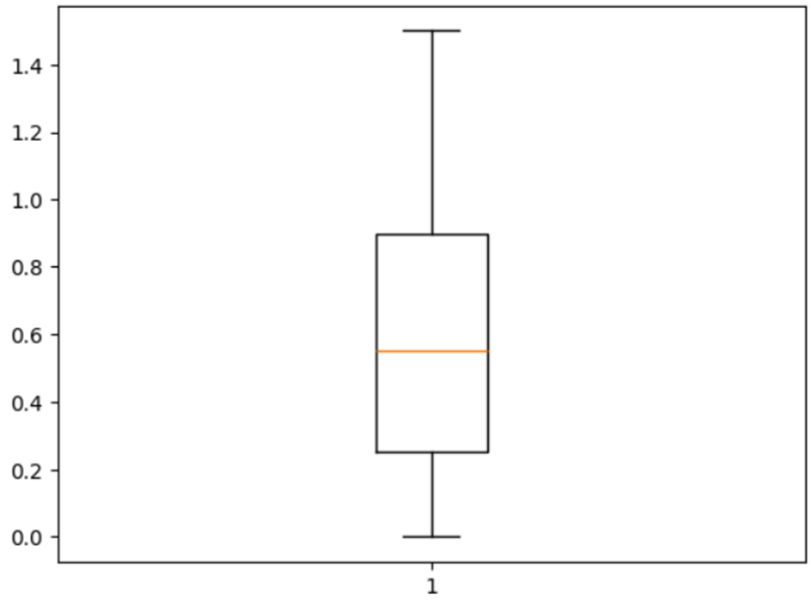
```
[7]: illness
     1    1638
     0    1554
     2     946
     3     542
     4     274
     5     236
     Name: count, dtype: int64
```

```
[8]: df["gender"].value_counts()
```

```
[8]: gender
     female    2702
     male      2488
     Name: count, dtype: int64
```

4) Visualize and analyze the maximum,mimimum and medium income

[9]:
```
y = list(df.income)
plt.boxplot(y)
plt.show()
```



5) Find out the no of days of reduced activity of male and female separately due to illness

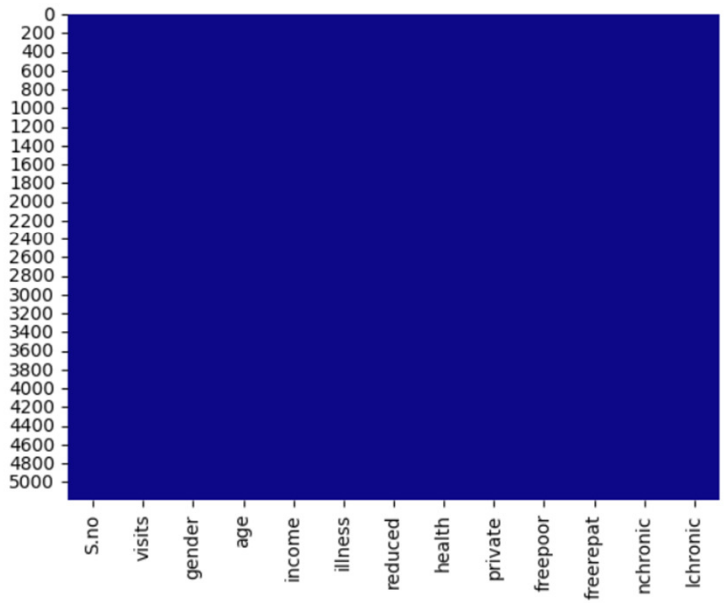
```
[13]: df.groupby('gender')['reduced'].mean()
```

```
[13]: gender
      female    0.975574
      male      0.738344
      Name: reduced, dtype: float64
```

6) Visualize is there is any missing values in the dataset based on a heatmap

```
[15]: import matplotlib.pyplot as plt
      import seaborn as sns
      sns.heatmap(df.isnull(),cbar=False,cmap='plasma')
```

```
[15]: <Axes: >
```



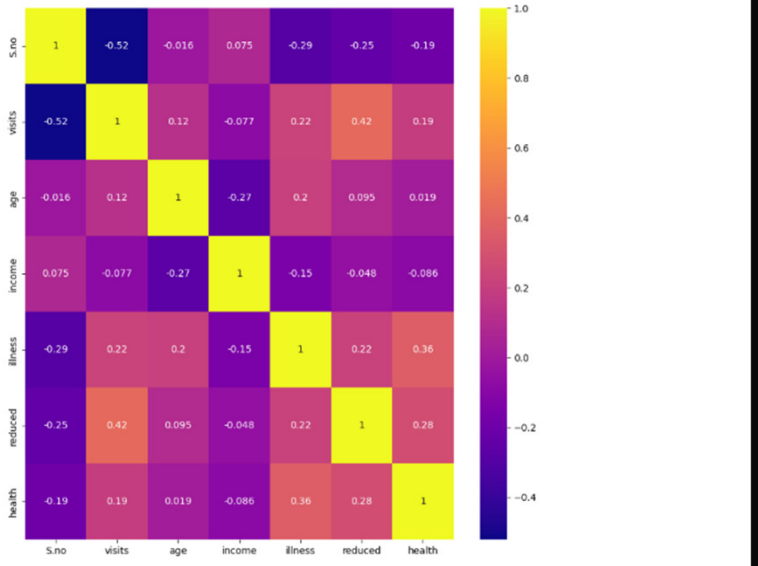
7) Find out the correlation between variables in the given dataset correlation between different variables

```
[23]: import matplotlib.pyplot as plt
      import seaborn as sns

      plt.figure(figsize=(10, 10))

      sns.heatmap(
          df.corr(numeric_only=True),
          cbar=True,
          annot=True,
          cmap='plasma'
      )

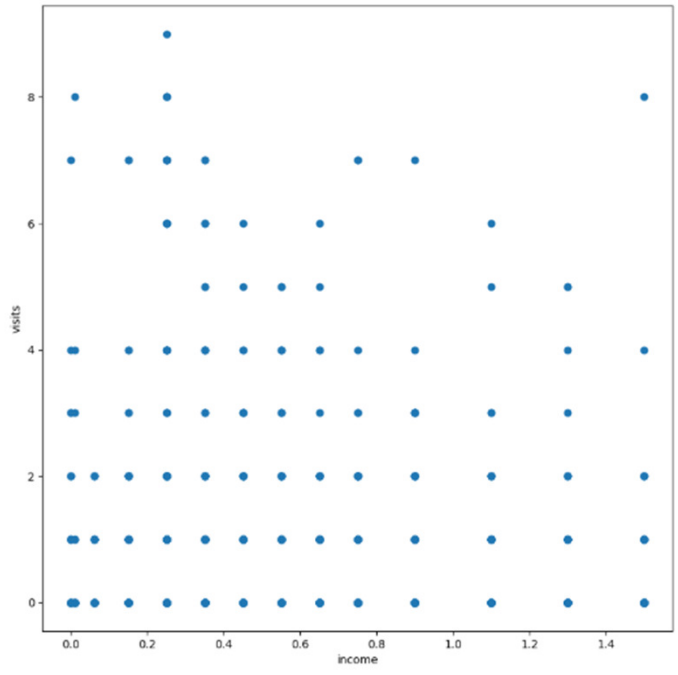
      plt.show()
```



8) Analyze how the income of a patient affects the no of visits to the hospital

```
[30]: import matplotlib.pyplot as plt
      import seaborn as sns
      plt.figure(figsize=(10,10))
      plt.scatter(x='income',y='visits',data=df)
      plt.xlabel('income')
      plt.ylabel('visits')

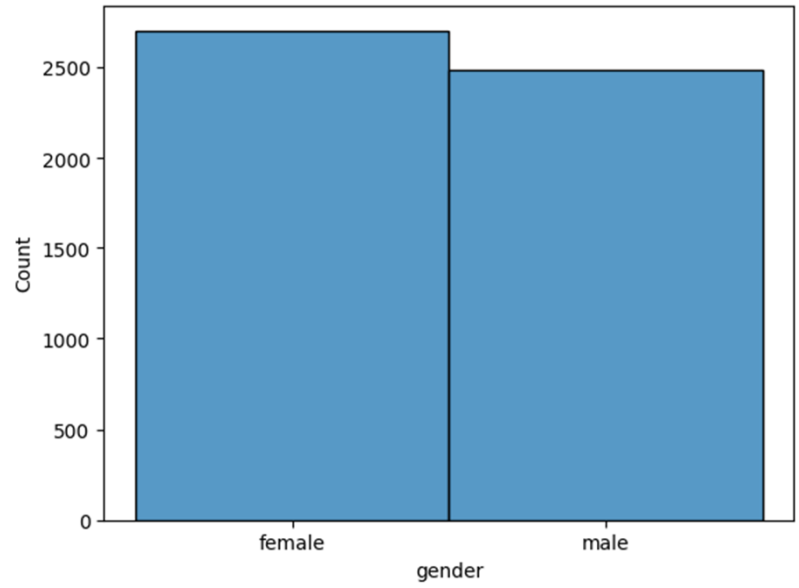
[30]: Text(0, 0.5, 'visits')
```



9) Count and visualize the number of males and females affected by illness

```
[44]: import matplotlib.pyplot as plt
      import seaborn as sns
      sns.histplot(df.gender,bins=2)
```

```
[44]: <Axes: xlabel='gender', ylabel='Count'>
```



10) Visualize the percentage of people getting govt health insurance due to low income, due to old age and also the percentage of people having private health insurance

[32]:
```
# % of People getting govt Insurance due to low income
label=['yes','no']
Y = df[df['freepoor']=='yes']
N = df[df['freepoor']=='no']
x = [Y.shape[0],N.shape[0]]
plt.figure(figsize=(5,5))
plt.pie(x,labels=label)
plt.title("% of people getting govt health Insurance due to low income")
plt.show()
# % of people having private Insurance
Y = df[df['private']=='yes']
N = df[df['private']=='no']
x = [Y.shape[0],N.shape[0]]
plt.figure(figsize=(5,5))
plt.pie(x,labels=label)
plt.title("% of people having private health Insurance")
plt.show()
# % of people getting govt Insurance due to old age, disability or veteran status
Y = df[df['freerepat']=='yes']
N = df[df['freerepat']=='no']
x = [Y.shape[0],N.shape[0]]
plt.figure(figsize=(5,5))
plt.pie(x,labels=label)
plt.title("% of people getting govt Insurance due to old age, disability or veteran status")
plt.show()
```

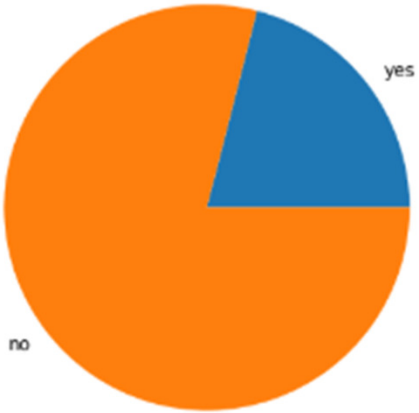
% of people getting govt health Insurance due to low income



% of people having private health Insurance

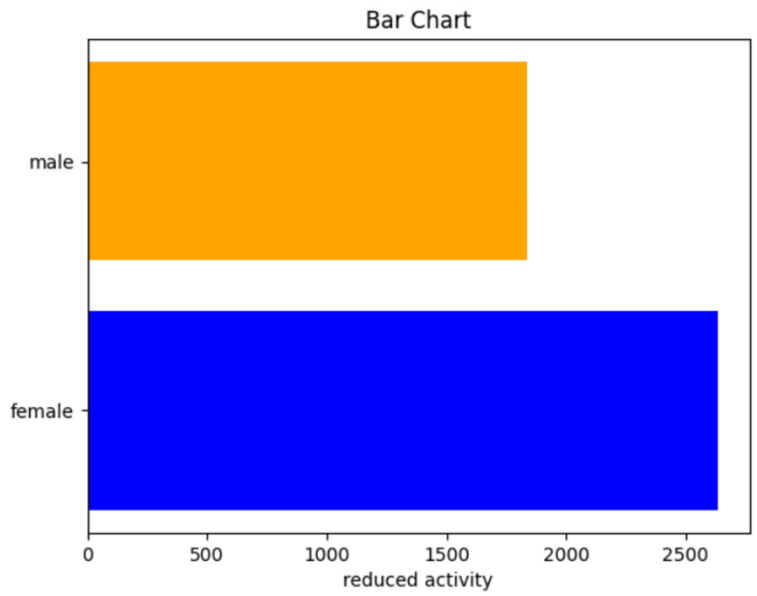


% of people getting govt Insurance due to old age, disability or veteran status



11) Plot a horizontal bar chart to analyze the reduced days of activity due to illness based on Gender

```
[40]: db= df.groupby('gender')['reduced'].sum().to_frame().reset_index()
      #Creating the bar chart
      plt.barh(db['gender'],db['reduced'],color = ['blue','orange'])
      plt.title('Bar Chart')
      plt.xlabel('gender')
      plt.xlabel('reduced activity')
      #Show the plot
      plt.show()
```



Future Scope

The Healthcare Analysis project has immense future scope as the healthcare organizations is constantly improving their use of data analytics for better patient care, reducing costs and supporting their decision making process in a data driven manner. Advanced technologies and use of additional health data can be used to enhance the project.

1. Predictive Healthcare Analytics - ML models can be built to predict health visits, chances of having a disease and a risk score for developing chronic illness based on age, income, health state and past healthcare use.

2. Early Disease Detection - Advanced analytics can be built upon medical data to help to detect potential risk of chronicity, medical conditions, chronic disease and other related issue at early stage.

3. Real time health dashboard - Interactive dashboard, such as with Power BI or Tableau, could be built for monitoring health visits, illness trends, chronic conditions and patient Demographics in real time.

4. Larger datasets - It can be expanded by integrating with medical records, admission at hospitals, treatment record and other medical data sets (medicine etc.) along with spatial data.

5. Automated healthcare report - Reports can be automatically generated on a regular basis for healthcare authorities which summarize health trends, visits of people, chronic diseases or other health indicators.

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Conclusion

The Healthcare Analysis task aims to extract significant trends and correlations on healthcare data among variables such as healthcare visits, age, gender, income, illness, health general and chronic conditions. Using cleaning techniques, Exploratory Data Analysis (EDA), statistical modeling and visualizing using Python, the dataset have been converted to insightful and actionable information. The results suggest trends on how demographic factors such as age, gender and income, illness, low activity, and poor health condition influence access to care. Moreover, health status and Chronic Conditions impact healthcare visits, while these relationships can be easily discovered using exploratory visualizations and analysis.

Thank You

Github Link

https://github.com/krishnamohanb21-svg/Healthcare-Analysis-DIY-Project-2-.git